
CSE (DATA SCIENCE)

Major Project : Phase-I
Year & Sem : IV & I Sem

A.Y : 2026-27
Date : 07.07.2026

Title of the project:

Enhanced AI-Based Human Activity Recognition with Fall Detection and Emergency Alert System

Project Batch No: B-08

S.No	H.T.No.	Name	Mail id (personal)	Mobile No.	Sign
01.	23K91A6797	K Nandini Reddy	kuntanandinireddy@gmail.com	9390821224	
02.	23K91A6787	K Mallishwari	kethavathmallishwari5@gmail.com	9391670647	
03.	23K91A6781	K Vivek	vivekvivek55544@gmail.com	9390294169	
04.	23K91A67B4	M Sagar	sagaryadav8443@gmail.com	8639236249	

Guide Name : **Mr. M. Arokia Muthu**

Designation : **Assistant Professor**

Mobile No : **9943436555**

Mail id : **arokyamuthu@tkrcet.com**

Abstract:

Human Activity Recognition (HAR) is an important application of Artificial Intelligence (AI) that enables the automatic identification of human activities for applications such as healthcare, elderly care, smart homes, surveillance, and human-computer interaction. Existing HAR systems mainly focus on recognizing daily activities but are limited by small datasets, controlled environments, and the absence of fall detection and emergency response mechanisms, reducing their effectiveness in real-world scenarios. The proposed **Enhanced AI-Based Human Activity Recognition with Fall Detection and Emergency Alert System** uses vision-based data to accurately recognize activities such as walking, standing, sitting, drinking water, and object picking. The system employs Artificial Intelligence and Deep Learning techniques with a larger and more diverse dataset to improve accuracy, robustness, and generalization under different environmental conditions, including low-light and partial occlusion scenarios. In addition, an AI-based Fall Detection module detects accidental falls in real time and automatically generates emergency alerts through SMS, email, mobile notifications, or alarms, enabling immediate assistance from caregivers or family members. The proposed system provides reliable real-time healthcare monitoring, enhances user safety, improves activity recognition performance, and offers a practical, scalable, and intelligent solution for elderly care, smart homes, and other real-world monitoring applications.

Keywords: Human Activity Recognition (HAR), Artificial Intelligence (AI), Deep Learning, Vision-Based Activity Recognition, Fall Detection, Emergency Alert System, Healthcare Monitoring, Elderly Care, Smart Homes.

Project Coordinator

Project Guide